Great Plains Population Flux

A Cellular Automaton Model

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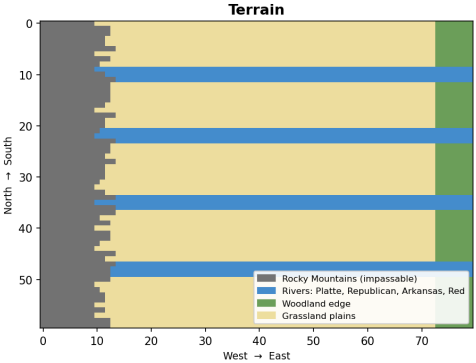
Overview

What we're modelling

- Two populations on a 2-D grid of the Great Plains:
  - Wildlife — bison herds
  - Humans — westward-moving settlers
- Shared resource pool (grass & water) per cell
- Discrete time steps; all updates fully vectorised (no per-agent loops)

Four flux drivers

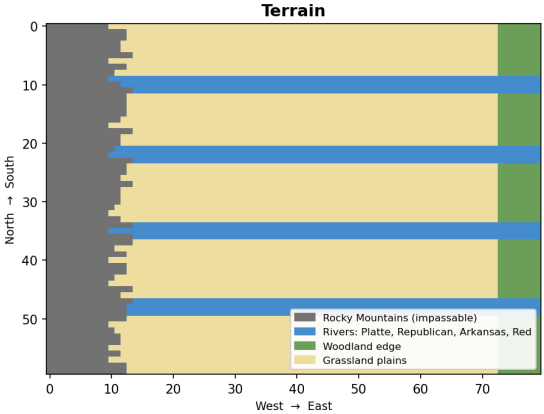
1. Resource gradients — move toward richer cells
2. Carrying-capacity pressure — overcrowded cells shed population
3. Climate events — stochastic drought & fire
4. Social attractors — rivers & towns pull settlers



The Grid & Terrain

Grid: 60×80 cells
Row 0 = North Col 0 = West

Terrain	Notes
Mountains	Western ~12 cols; impassable barrier; jagged edge (± 2 cols/row)
Rivers	4 E-W bands at fixed rows; higher resource cap (1.4 vs 1.0); approximate Platte, Republican, Arkansas, Red
Woodland	Eastern ~7 cols; lower resource cap (0.75); settlers' entry point
Plains	Remainder; standard resource cap (1.0)



Model State

Four 2-D arrays evolve each step:

Layer	Units	Description
`resources`	$[0, C]$	Combined grass/water abundance; C varies by terrain
`wildlife`	count	Bison population density; starts as Gaussian blob centred in plains
`humans`	count	Settler population density; starts along the eastern woodland edge
`infra`	$[0, 1]$	Infrastructure (trails, towns); grows with human presence, decays when settlers leave

Every array is multiplied by a static **mask** (0 on mountains, 1 elsewhere) after each update, so mountains stay permanently empty.

The Flux Mechanism — Directional Weights

For each cell (i, j) and each cardinal direction $d \in \{N, S, E, W\}$:

Attractiveness & softmax weights

$$a_d(i,j) \;=\; \text{attractiveness of the neighbour in direction } d$$

$$w_d \;=\; \exp\!\big(\beta \cdot a_d\big) \cdot \mathbf{1}[\text{neighbour is passable}]$$

$$f_d \;=\; \frac{w_d}{\sum_{d'} w_{d'}}, \qquad \sum_d f_d = 1$$

- $\beta =$ `drift_strength`: large $\beta \Rightarrow$ decisive gradient-following; small $\beta \Rightarrow$ diffusive random walk
- Invalid neighbours (mountains, grid boundary) get $w_d = 0$, so population never flows into them
- Normalisation to 1 **guarantees mass conservation** during movement
- Attractiveness for *wildlife*: resources only

The Flux Mechanism — Outflow & Inflow

Total outflow from cell (i,j)

$$\mathrm{out}(i,j) \;=\; \min\Big(P,\; r_b\,P \;+\; r_p\,\max\big(0,\; P-K\big)\Big)$$

P	local population
r_b	`base_flux` — baseline restlessness each step
r_p	`pressure_rate` — extra outflow per unit above K
K	`carrying_cap` — pressure threshold (not a hard ceiling)

Directed outflow: $\mathrm{out}_d = \mathrm{out} \cdot f_d$

Inflow at (i,j) = sum of directed outflows from each of the four neighbours pointing toward (i,j), computed by shifting the outflow arrays by one cell.

Net delta $= \mathrm{inflow} - \mathrm{outflow}$, then multiplied by mask.

Other Per-Step Dynamics

Resource regeneration (logistic)

$$\Delta R = r_{\text{regen}} \cdot R \cdot \left(1 - \frac{R}{C}\right)$$

Fast recovery when depleted; slows near cap C.

Climate events (per step)

- Drought: prob. 0.025, radius 9, severity 0.55
- Fire: prob. 0.012, radius 5, severity 0.70
- Epicentre chosen at random; Gaussian damage footprint
- Resources multiplied by $(1 - \text{damage})$

Logistic birth / death

$$\Delta P_+ = r_+ \cdot P\left(1 - \frac{P}{K}\right), \quad \Delta P_- = r_- \cdot P$$

Consumption (grazing / settlement)

$$\Delta R_{\text{consumed}} = \min\big(R,\ P \cdot c\big)$$

Bison graze *before* settlers consume, so they compete for the same pool within a step.

Infrastructure

$$\Delta I = g\,\frac{P}{K} - d\,I\left(1 - \frac{P}{K}\right)$$

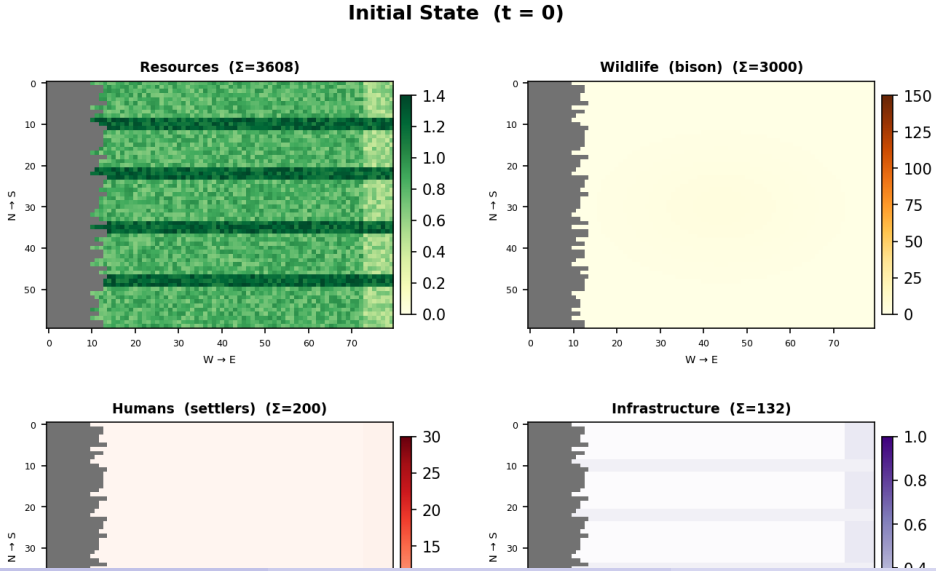
Update Order Each Step

Order matters — populations respond to the *current* landscape:

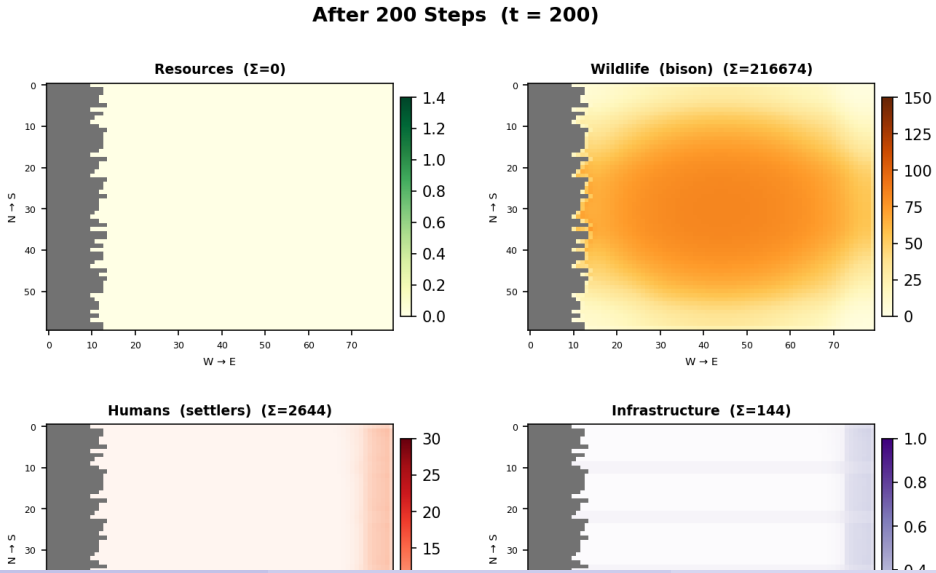
1. **Resource regeneration** — logistic recovery toward terrain cap
2. **Climate event** — stochastic drought or fire (or neither)
3. **Wildlife update** — flux + birth/death + grazing
4. **Human update** — flux + birth/death + consumption
   (sees resources already depleted by bison)
5. **Infrastructure update** — towns grow/decay based on human density
6. **Record snapshot** (every `record_every` steps)

All state arrays are updated **synchronously** from the previous step's values — standard CA practice that avoids order-of-operations artefacts within a step.

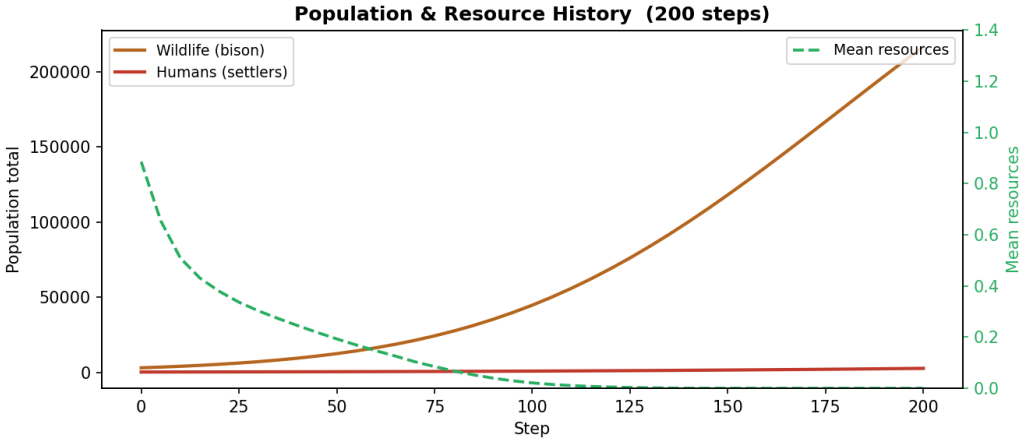
Initial State ($t = 0$)



After 200 Steps ($t = 200$)



Population & Resource History



Bison outpace the grassland's recovery rate and crash resources by ∼step 75.
Settlers remain sparse — their slower dynamics are masked by the bison explosion.

Known Dynamics & Next Steps

Current behaviour (default params)

- Bison undergo exponential growth then resource crash by ∼step 75
- Settlers grow slowly; swamped by bison in resource competition
- Infrastructure seeds along rivers and woodland edge but spreads slowly

Tuning levers for stable dynamics

- Lower `wl_consumption` (less aggressive grazing)
- Raise `res_regen_rate` (faster grassland recovery)
- Lower `wl_carrying_cap` (earlier pressure to disperse)

Planned extensions

- Wildlife–human repulsion (bison flee settler-dense cells)
- Seasonal resource variation (modulate regrowth rate by season)
- Historical scenario mode (introduce railroads, hunting pressure at specific steps)
- Spatially correlated initial resources (gaussian-blurred noise)
- Finer inline comments inside `_flux()` method

Code

Single file: `plains_ca.py`